



Improved speech emotion recognition with Mel frequency magnitude coefficient

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Abstract

Automatic speech emotion recognition using machine learning is a demanding research topic in the field of affective computing. Identifying the speech features for speech emotion recognition is a challenging issue as the feature needs to emphasize the information about emotion from the speech. Spectral features play a vital role in emotion recognition from speech signals. In this paper, two modifications are made in the extraction of Mel frequency cepstral coefficient, they are, using magnitude spectrum instead of energy spectrum and exclusion of discrete cosine transform and extract Mel Frequency Magnitude Coefficient. Mel frequency magnitude coefficient is the log of magnitude spectrum on a non-linear Mel scale frequency. Mel frequency magnitude coefficient and three conventional spectral features, Mel frequency cepstral coefficient, log frequency power coefficient and linear prediction cepstral coefficient are tested on Berlin, Ravdess, Savee, EMOVO, eINTERFACE and Urdu databases with multiclass support vector machine as the classifier. Mel frequency magnitude coefficient as a stand alone feature recognizes emotion with an accuracy of 81.50% for Berlin, 64.31% for Ravdess, 75.63% for Savee, 73.30% for EMOVO, 56.41% for eINTERFACE and 95.25% for Urdu databases. Mel frequency magnitude coefficient is found to be the better spectral feature for the identification of emotion from speech compared to the conventional features.

Introduction

Speech is an exceptional characteristic of human which conveys information from one person to another. Emotion is the reaction of humans to a particular situation that can be expressed in words or deeds. Every person expresses his/her emotions differently for the identical situation. Because of

this reason, automatic speech emotion recognition (SER) is an effective research topic especially in emotionally rich man-machine interaction system. This research area makes the system to recognize and understand the emotions of human beings for wide variety of purposes. Some of the applications of SER are service robots to recognize negative and non-negative emotions of user to provide service accordingly [1], surveillance systems by identifying fear-type emotions [2], call centers to find angry customers [3], deception detection [4], etc.

The Mel frequency cepstral coefficient (MFCC) is the feature mostly used in automatic speech emotion recognition systems. MFCC is the discrete cosine transform of log of power spectrum on a Mel scale frequency. MFCC represents speech in a better way by taking the advantage of auditory perception of humans and it is used in most of the speech emotion recognition studies. But the research community is yet to attain better emotion recognition rate using MFCC for all type of emotions and databases. It could be because of the following reasons: i) Short time energy is used to extract MFCC, energy is not a satisfactory feature for large signal level as it uses square function [5], ii) The transformation vector of discrete cosine transform (DCT) includes all frequencies. If a frequency band is corrupted by any noise factor, it affects all the MFCCs [6] and iii) If voiced phoneme is adjacent to unvoiced phoneme in a frame, the dominant voiced phoneme will be available in the frame and unvoiced phoneme will vanish [6]. This leads to information loss in the speech. Therefore, it is necessary to find a better spectral feature to effectively identify different emotions from speech signals spoken by people of different language, culture etc. In this paper, we have modified the extraction process of MFCC to overcome the above mentioned problems and extracted Mel frequency magnitude coefficient (MFMC) which recognizes the emotions with better accuracy than the traditional spectral features including MFCC and log frequency power coefficient (LFPC).

The forthcoming sections are arranged as follows: Section 2 discusses the work related to speech emotion recognition, Section 3 illustrates the extraction of Mel frequency magnitude coefficient, Section 4 explains the databases, extraction of standard spectral features which are used for comparative analysis and experimental validation of speech emotion recognition, Section 5 discusses the emotion recognition performance of standard spectral features and the MFMC feature and Section 6 presents the conclusion and future enhancement.

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Related works

Speech emotion recognition interprets speaker's emotional state from their speech using four phases: emotional speech data collection, speech feature extraction, classification and performance

evaluation. The first step in emotion recognition system is the collection of databases. The classification accuracy relies on the naturalness of the database. Speech databases are categorized into three: acted, induced and natural. In acted database, the primary emotions are acted by the professional ...

Mel Frequency Magnitude Coefficient

This section explains extraction of the MFMC and the modifications in the process of extraction of MFMC in comparison with the traditional MFCC. MFMC is extracted the same way as that of Mel frequency cepstral coefficient with the exception of two steps: first, magnitude of fast Fourier transform is used instead of magnitude square. Second, the discrete cosine transform used in the MFCC extraction for the purpose of decorrelation is excluded. The processes involved in the extraction of MFCC and ...

Emotional speech corpuses

Six databases are used in this paper; they are, Berlin database [38], Ravdess database [39], Savee database [40], EMOVO emotional speech database [41], eINTERFACE database [42] and Urdu database [43].

In Berlin database [38], ten German utterances, 5 long and 5 short sentences in seven basic emotions such as anger, boredom, disgust, fear, happy, neutral and sadness have been recorded by 10 professional actors, 5 males and 5 females. The recordings have been done in anechoic chamber of Technical ...

Results and discussions

The accuracies of emotion recognition from speech signal using LPCC, LFPC, MFCC and the MFMC features with multiclass SVM classifier are shown in Table 2. The MFMC feature performs better than the other three features in recognizing emotions of all the six databases.

For Berlin database, MFMC recognizes emotions with an accuracy of 81.50% which is the highest among the recognition accuracies of all other features. MFMC recognizes the emotions with improved accuracy of 7.29% higher than LFPC, ...

Conclusion

In this paper, Mel frequency magnitude coefficient has been used for speech emotion recognition. Mel frequency magnitude coefficient differs from Mel frequency cepstral coefficient in two aspects; first it uses magnitude instead of energy and second it excludes discrete cosine transform. Experimental results show that the Mel frequency magnitude coefficients classify emotions better than Mel frequency cepstral coefficients, log frequency power coefficients and linear predictive cepstral ...

CRedit authorship contribution statement

J. Ancilin: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft. **A. Milton:** Methodology, Software, Validation, Resources, Supervision. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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Citation Excerpt :

...They evaluated the system on RAVDESS and Toronto emotional speech set (TESS) database using three classifiers namely decision tree (DT), CNN, SVM, and attained maximum accuracy of 96% for CNN with TESS dataset. A study used a magnitude spectrum for extraction of Mel frequency cepstral coefficient (MFCC) without discrete cosine transform (DCT) [56]. The evaluation of this study was performed on six datasets namely Berlin, RAVDESS, SAVEE, EMOVO, eINTERFACE, and Urdu datasets....

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...The proposed system was evaluated on different datasets including IEMOCAP, EMO-DB, eINTERFACE, SAVEE and achieved 71.98%, 82.10%, 75.60%, 54.75% accuracy scores respectively. Ancilin et al. [18] extracted the Mel Frequency Magnitude Coefficient which was the log of magnitude spectrum on a non-linear Mel scale

frequency. The authors showed that the Mel frequency magnitude coefficient was a better spectral feature to identify emotion from speech compared to the conventional features....

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